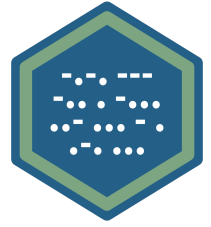


Codebusters

(Scheduled Event - 30 Minutes)



Description: Teams will use cryptanalysis techniques to decrypt encrypted messages using a variety of historical and modern ciphers.

Participants per assigned Team Number: 3

- If your school has 1 team you will send 3 students
- If your school has 2 teams you will send 3 students per team number; team numbers may not intermix
- If your school has 3 teams you will send 3 students per team number; team numbers may not intermix

Teams must bring:

- Pen/Pencil

Teams may bring:

- Four-function calculator
- [VASO Division A Codebuster Resources Sheet](#)

Teams may not bring:

- Any notes or guides of any kind.

Safety Requirements:

- None

Materials Provided at the Event:

- Work space sheets and scratch paper
- [VASO Division A Codebusters Resources Sheet](#)

The Competition: This event consists of a written exam in which participants use cryptanalysis techniques to decrypt encrypted messages using one or more of the following cipher types:

(No code will have a readability higher than 9th grade. Some ciphers are more difficult and will therefore have a higher point total possible, e.g. Caesar is easier than Aristocrats, so it will have a higher possible point designation.)

The code types that may be used for this exam are:

- Aristocrats with a hint (Messages with spaces included, and with the hint; no letter can ever decrypt to itself)
- Atbash Cipher (English alphabet)
- Caesar (Shift) cipher
- PigPen/Masonic
- Polybius Square
- Running Man
- Vigenère Cipher (with provided key)

Bonus Decoding:

- Simple Cryptarithm

Scoring:

- The high score wins.
- Scores are calculated on the total number of correct ciphers solved.
- Each complete and correct cypher earns a bonus 1/10 of the total cypher characters.

Tie Breakers:

- Ties are broken by review of the worksheets showing that all characters are in the correct placement for each cipher.

Resources:

TOEBES: [Science Olympiad Code Busters](#)

Allowed Calculator

This is the acceptable calculators for all Division A events.

Class I - Stand-alone non-graphing, non-programmable, non-scientific 4-function or 5-function calculators. These are the most basic types of calculators. They are limited to the four basic mathematical functions and square roots. They are often cheap, costing as low as a dollar online.



Safety Glasses

Indirect Vent Goggles or Safety glasses are a mandatory safety requirement for some events. It is important to choose the type of protection specific to the task. Competitors are responsible for providing their own protective eyewear. Teams will not be allowed to compete without adequate eye protection.

Category C: Indirect Vent Goggles



Category B: Safety glasses



Compliant Eyewear Categories:

CATEGORY B

- Description: Impact protection. They provide protection from a high inertia particle hazard (high mass or velocity)
- Corresponding ANSI designation/required marking: Z87+.
- Category B eyewear can look either like goggles or safety glasses, but must have the Z87+ marking embossed on them.

CATEGORY C

- Description: Indirect vent chemical/splash protection goggles. These seal tightly to the face completely around the eyes and have indirect vents constructed so that liquids do not have a direct path into the eye (or no vents at all). If you can see through the vent holes from one side to the other, they are NOT indirect vents
- Corresponding ANSI designation/required marking: Z87 (followed by D3 is the most modern designation but, it is not a requirement)

Examples of Non-Compliant Eyewear:

- Face shields/visors are secondary protective devices and are not approved.
- Prescription Glasses containing safety glass should not be confused with safety spectacles. "Safety glass" indicates the glass is made to minimize shattering when it breaks. Unless these glasses bear the Z87+ mark they are not approved for use.

Notes:

1. A goggle that bears the Z87+ mark and is an indirect vent chemical/splash protection goggle will qualify for both Category B & C
2. VisorGogs do not seal completely to the face, but are acceptable as Type C goggles